

PHYS 121 — HW6 Solutions

Juntang Wang

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Q1. Temperature of a pendulum (4+4 pts)

Period of a simple pendulum (small oscillations):

$$T = 2\pi\sqrt{\frac{L}{g}} \quad \Rightarrow \quad \frac{dT}{T} = \frac{1}{2} \frac{dL}{L}.$$

Thermal expansion: $dL = \alpha L dT$, so

$$\frac{dT_{\text{period}}}{T} = \frac{1}{2} \frac{dL}{L} = \frac{1}{2} \alpha dT.$$

If the clock is correct at T_{cal} and the room is at $T_{\text{room}} = T_{\text{cal}} + \Delta T$ with $\Delta T > 0$, then L increases, so T increases and the clock *runs slow* (falls behind).

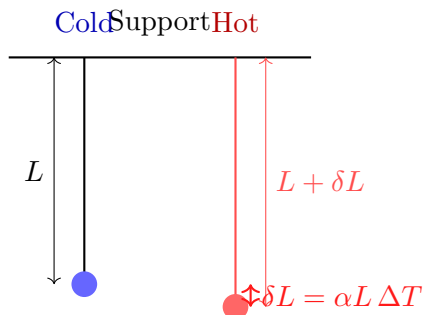
Over one true day $t_{\text{day}} = 86400$ s, the fractional error in the time kept by the clock is the same:

$$\frac{\Delta t}{t_{\text{day}}} = \frac{1}{2} \alpha \Delta T \quad \Rightarrow \quad \Delta t = \frac{1}{2} \alpha \Delta T t_{\text{day}}.$$

Since T is larger, this is a *loss* of time:

$$\Delta t_{(\text{slow})} = \frac{1}{2} \alpha \Delta T (86400 \text{ s/day}).$$

Required sketch:



Q2. Estimate Earth's temperature (2+3+3+1 pts)

Solar constant at Earth's orbit: $S \simeq 1370 \text{ W/m}^2$. Stefan-Boltzmann constant: σ .

(a) Intercepted power

The Earth intercepts sunlight over its geometric cross section:

$$P_{\text{in}} = S \pi R^2.$$

If it is a blackbody at temperature T , it radiates from its whole surface area $4\pi R^2$:

$$P_{\text{out}} = 4\pi R^2 \sigma T^4.$$

Equating $P_{\text{in}} = P_{\text{out}}$ gives

$$S \pi R^2 = 4\pi R^2 \sigma T^4 \quad \Rightarrow \quad T = \left(\frac{S}{4\sigma} \right)^{1/4} \approx \boxed{279 \text{ K}}.$$

(b) Including albedo

With albedo $A = 0.3$, only a fraction $(1 - A)$ is absorbed:

$$P_{\text{in}} = (1 - A) S \pi R^2.$$

Set equal to $4\pi R^2 \sigma T^4$:

$$(1 - A) S \pi R^2 = 4\pi R^2 \sigma T^4 \quad \Rightarrow \quad T = \left(\frac{(1 - A) S}{4\sigma} \right)^{1/4} \approx \boxed{255 \text{ K}}.$$

(c) Role of greenhouse effect

Observed mean surface temperature is about 288 K, well above 255 K. Thus atmospheric greenhouse warming is important:

yes, the greenhouse effect is significant

.

(d) Dependence on Earth's area

Both absorbed and emitted powers scale as R^2 , so they cancel in the balance. Therefore

T is independent of Earth's radius (area) in this model.

Q3. Unreasonable results? H₂S speeds (3+5 pts)

Molar mass of H₂S: $M = 31.4 \text{ g/mol} = 0.0314 \text{ kg/mol}$.

(a) Average speed

For an ideal gas, the mean speed is

$$\bar{v} = \sqrt{\frac{8RT}{\pi M}}.$$

At $T = 250 \text{ K}$,

$$\bar{v} = \sqrt{\frac{8(8.314)(250)}{\pi(0.0314)}} \approx \boxed{4.1 \times 10^2 \text{ m/s}}.$$

(b) Reliability of this result

At 250 K and realistic pressures, H_2S is far from an ideal, nearly noninteracting gas: it has strong intermolecular attractions and a relatively high boiling point. The assumptions of the Maxwell-Boltzmann distribution (point particles, negligible interactions except elastic collisions, ideal equation of state) are poor. Thus the numerical speed is much less reliable than for ideal, monoatomic gases; real interactions and non-ideality matter.

Q4. Airtight water dispenser (10 pts)

Given: Cross-section $A = 25 \text{ cm} \times 10 \text{ cm} = 0.025 \text{ m}^2$, total height $H = 20 \text{ cm} = 0.20 \text{ m}$, initial water level $h_0 = 20 - 3.5 = 16.5 \text{ cm} = 0.165 \text{ m}$, initial air pressure $P_{0,\text{air}} = 1.00 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$, water density $\rho = 1000 \text{ kg/m}^3$, temperature constant (isothermal).

(a) Pressure at bottom

$$P_{\text{bottom}} = P_{\text{air}} + \rho gh.$$

For gauge pressure = 0, $P_{\text{bottom}} = P_{\text{atm}} = 1.00 \text{ atm}$.

(b) Isothermal air relation

For isothermal air: $P_{\text{air}} V_{\text{air}} = \text{constant}$.

Initial: $V_{0,\text{air}} = A(H - h_0) = A(0.20 - 0.165) = 8.75 \times 10^{-4} \text{ m}^3$.

Final: $V_{\text{air}} = A(H - h)$, so

$$P_{\text{air}} = P_{0,\text{air}} \frac{V_{0,\text{air}}}{V_{\text{air}}} = P_{0,\text{air}} \frac{H - h_0}{H - h}.$$

(c) Final water height and volume

At equilibrium: $P_{\text{bottom}} = P_{\text{air}} + \rho gh = P_{\text{atm}}$.

Substituting:

$$P_{0,\text{air}} \frac{H - h_0}{H - h} + \rho gh = P_{\text{atm}}.$$

With $P_{0,\text{air}} = P_{\text{atm}}$, this simplifies to:

$$P_{\text{atm}} \frac{H - h_0}{H - h} + \rho gh = P_{\text{atm}}.$$

Rearranging:

$$P_{\text{atm}} \left(\frac{H - h_0}{H - h} - 1 \right) = -\rho gh,$$

$$P_{\text{atm}} \frac{h - h_0}{H - h} = -\rho gh.$$

Multiplying both sides by $(H - h)$:

$$P_{\text{atm}}(h - h_0) = -\rho gh(H - h).$$

Rearranging: $h_0 = h + \frac{(H-h)\rho gh}{P_{\text{atm}}}$.

This gives a quadratic in h :

$$\frac{\rho g}{P_{\text{atm}}} h^2 - \left(1 + \frac{H \rho g}{P_{\text{atm}}}\right) h + h_0 = 0.$$

Using $P_{\text{atm}} = 1.013 \times 10^5 \text{ Pa}$, $\rho g = 9800 \text{ N/m}^3$, $H = 0.20 \text{ m}$, $h_0 = 0.165 \text{ m}$:

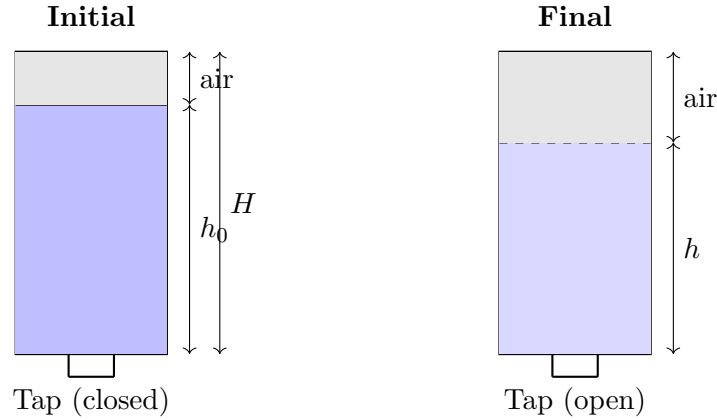
$$0.0967h^2 - 1.0193h + 0.165 = 0.$$

Solving: $h = 0.1644 \text{ m} = 16.4 \text{ cm}$.

Volume of water flowed out:

$$\Delta V = A(h_0 - h) = 0.025 \times (0.165 - 0.1644) = \boxed{1.5 \times 10^{-5} \text{ m}^3 = 15 \text{ cm}^3}.$$

Required drawing:



Q5. Adiabatic process (3+3+3+3 pts)

Given: $n = 1 \text{ mol}$, $V_1 = 1.0 \times 10^{-2} \text{ m}^3$, $P_1 = 2.0 \times 10^5 \text{ N/m}^2$, $V_2 = 5.0 \times 10^{-3} \text{ m}^3$, monatomic gas ($\gamma = 5/3$), adiabatic compression.

(a) Initial temperature

From ideal gas law: $P_1 V_1 = n R T_1$,

$$T_1 = \frac{P_1 V_1}{n R} = \frac{2.0 \times 10^5 \times 1.0 \times 10^{-2}}{1 \times 8.314} = \boxed{241 \text{ K}}.$$

(b) Final temperature

For adiabatic process: $T V^{\gamma-1} = \text{constant}$, so

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 241 \left(\frac{1.0 \times 10^{-2}}{5.0 \times 10^{-3}} \right)^{2/3} = 241 \times 2^{2/3} \approx \boxed{382 \text{ K}}.$$

(c) Work done on gas

For adiabatic process: $W = \frac{nR}{\gamma-1}(T_1 - T_2)$,

$$W = \frac{1 \times 8.314}{5/3 - 1}(241 - 382) = \frac{8.314}{2/3} \times (-141) = 1.76 \times 10^3 \text{ J}.$$

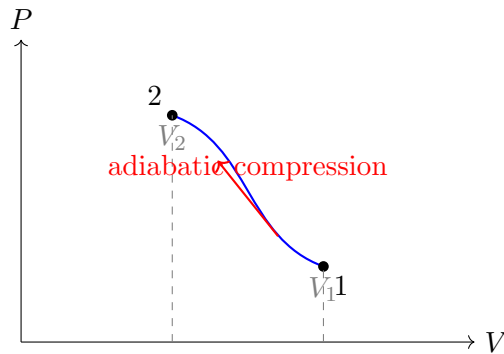
Work done *on* the gas is $\boxed{+1.76 \times 10^3 \text{ J}}$.

(d) Change in internal energy

For monatomic ideal gas: $\Delta U = \frac{3}{2}nR(T_2 - T_1)$,

$$\Delta U = \frac{3}{2} \times 1 \times 8.314 \times (382 - 241) = \boxed{1.76 \times 10^3 \text{ J}}.$$

Required drawing:



Q6. Inelastic collision heating (5+5 pts)

Given: $m_b = 10 \text{ g} = 0.010 \text{ kg}$, $v_{b,0} = 200 \text{ m/s}$, $m_p = 2.0 \text{ kg}$, $C_V = 3R$, $M = 200 \text{ g/mol} = 0.200 \text{ kg/mol}$.

(a) Mechanical energy dissipated

Momentum conservation: $m_b v_{b,0} = (m_b + m_p) v_f$,

$$v_f = \frac{m_b v_{b,0}}{m_b + m_p} = \frac{0.010 \times 200}{2.01} = 0.995 \text{ m/s}.$$

Initial kinetic energy: $K_i = \frac{1}{2} m_b v_{b,0}^2 = \frac{1}{2} \times 0.010 \times 200^2 = 200 \text{ J}$.

Final kinetic energy: $K_f = \frac{1}{2} (m_b + m_p) v_f^2 = \frac{1}{2} \times 2.01 \times 0.995^2 = 0.996 \text{ J}$.

Energy dissipated:

$$\Delta E = K_i - K_f = 200 - 0.996 = \boxed{199 \text{ J}}.$$

(b) Temperature increase

Total mass: $m_{\text{tot}} = m_b + m_p = 2.01 \text{ kg}$.

Number of moles: $n = \frac{m_{\text{tot}}}{M} = \frac{2.01}{0.200} = 10.05 \text{ mol}$.

Heat capacity at constant volume: $C_V = 3R = 24.94 \text{ J/(mol} \cdot \text{K)}$.

Total heat capacity: $C_{V,\text{tot}} = nC_V = 10.05 \times 24.94 = 250.6 \text{ J/K}$.

Temperature increase:

$$\Delta T = \frac{\Delta E}{C_{V,\text{tot}}} = \frac{199}{250.6} = \boxed{0.794 \text{ K}}.$$

Q7. A Star is Born (3+3+3+2 pts)

Given: Initial radius R_i , initial density ρ_i , opacity density ρ_1 , initial temperature T_1 , adiabatic index $\gamma = 7/5$.

(a) Radius at opacity

Mass conservation: $M = \frac{4}{3}\pi R_i^3 \rho_i = \frac{4}{3}\pi r_1^3 \rho_1$,

$$r_1^3 = R_i^3 \frac{\rho_i}{\rho_1} \quad \Rightarrow \quad r_1 = R_i \left(\frac{\rho_i}{\rho_1} \right)^{1/3}.$$

(b) Temperature scaling and r_2

For adiabatic process: $TV^{\gamma-1} = \text{constant}$, so

$$Tr^{3(\gamma-1)} = \text{constant} \quad \Rightarrow \quad T \propto r^{-3(\gamma-1)} = r^{-6/5}.$$

At temperature T_2 :

$$T_2 r_2^{6/5} = T_1 r_1^{6/5} \quad \Rightarrow \quad r_2 = r_1 \left(\frac{T_1}{T_2} \right)^{5/6}.$$

(c) Numerical r_1

With $R_i = 3.0 \times 10^{15} \text{ m}$, $\rho_i = 2.0 \times 10^{-14} \text{ g/m}^3 = 2.0 \times 10^{-17} \text{ kg/m}^3$, $\rho_1 = 1.0 \times 10^{-7} \text{ g/m}^3 = 1.0 \times 10^{-10} \text{ kg/m}^3$:

$$r_1 = 3.0 \times 10^{15} \left(\frac{2.0 \times 10^{-17}}{1.0 \times 10^{-10}} \right)^{1/3} = 3.0 \times 10^{15} \times (2.0 \times 10^{-7})^{1/3}.$$

$$r_1 = 3.0 \times 10^{15} \times 5.848 \times 10^{-3} = \boxed{1.75 \times 10^{13} \text{ m}}.$$

(d) Numerical r_2

With $T_2 = 2000 \text{ K}$, and assuming T_1 is the initial temperature (typically around 10 K for molecular clouds):

$$r_2 = r_1 \left(\frac{T_1}{T_2} \right)^{5/6} = 1.75 \times 10^{13} \left(\frac{T_1}{2000} \right)^{5/6}.$$

If $T_1 = 10 \text{ K}$:

$$r_2 = 1.75 \times 10^{13} \times \left(\frac{10}{2000} \right)^{5/6} = 1.75 \times 10^{13} \times (0.005)^{5/6} \approx 1.75 \times 10^{13} \times 0.0121 \approx \boxed{2.1 \times 10^{11} \text{ m}}.$$

Q8. A process: piston + mass, adiabatic vs isothermal (3+4+4+3 pts)

Given: $n = 1$ mol He (monatomic, $\gamma = 5/3$), $T_1 = 27^\circ\text{C} = 300\text{ K}$, $P_1 = 100.0\text{ kPa} = 1.00 \times 10^5\text{ Pa}$, piston area $A = 0.0010\text{ m}^2$, mass on piston $m = 2.0\text{ kg}$, bath temperature $T_{\text{bath}} = 27^\circ\text{C} = 300\text{ K}$, and atmospheric pressure $P_{\text{atm}} = P_1$.

Initial volume:

$$V_1 = \frac{nRT_1}{P_1} = \frac{1 \times 8.314 \times 300}{1.00 \times 10^5} = 0.0249\text{ m}^3.$$

(a) Minimum volume during first dip

When the mass is suddenly placed on the piston, the gas is compressed adiabatically (rapid process, no heat exchange).

The work done on the gas comes from:

- The added mass: pressure mg/A over area A ,
- The atmosphere: pressure P_{atm} .

Total external pressure on the gas during the adiabatic compression is

$$P_{\text{ext}} = P_{\text{atm}} + \frac{mg}{A}.$$

Total work done *on* the gas as it goes from V_1 to V_{min} :

$$W_{\text{on}} = P_{\text{ext}}(V_1 - V_{\text{min}}).$$

For an adiabatic process, $Q = 0$ so $\Delta U = W_{\text{on}}$. For one mole of ideal gas,

$$\Delta U = \frac{nR}{\gamma - 1}(T_{\text{min}} - T_1),$$

and from the adiabatic relation $TV^{\gamma-1} = \text{const}$,

$$T_{\text{min}} = T_1 \left(\frac{V_1}{V_{\text{min}}} \right)^{\gamma-1}.$$

Putting this together:

$$\frac{nR}{\gamma - 1} \left[T_1 \left(\frac{V_1}{V_{\text{min}}} \right)^{\gamma-1} - T_1 \right] = \left(P_{\text{atm}} + \frac{mg}{A} \right) (V_1 - V_{\text{min}}).$$

Substituting $P_{\text{atm}} = 1.00 \times 10^5\text{ Pa}$, $mg = 19.6\text{ N}$, $A = 1.0 \times 10^{-3}\text{ m}^2$, $nR = 8.314\text{ J/K}$, $\gamma = 5/3$, $T_1 = 300\text{ K}$, $V_1 = 0.0249\text{ m}^3$, we solve numerically and find

$$V_{\text{min}} \approx \boxed{0.020\text{ m}^3}.$$

(b) Equilibrium volume after damping (adiabatic)

After some oscillations, the piston comes to rest mechanically, but the gas is still not in thermal equilibrium with the bath. At this mechanical equilibrium:

$$P_{\text{gas}} = P_{\text{atm}} + \frac{mg}{A} = 1.00 \times 10^5 + \frac{19.6}{0.0010} = 1.196 \times 10^5\text{ Pa}.$$

Between the initial state and this mechanically equilibrated state, the gas has undergone an adiabatic process, so

$$P_1 V_1^\gamma = P_{\text{gas}} V_{\text{eq,adiabatic}}^\gamma.$$

Thus

$$V_{\text{eq,adiabatic}} = V_1 \left(\frac{P_1}{P_{\text{gas}}} \right)^{1/\gamma} = 0.0249 \times \left(\frac{1.00 \times 10^5}{1.196 \times 10^5} \right)^{3/5} \approx \boxed{0.0224 \text{ m}^3}.$$

(c) Final equilibrium volume

After a long time, the gas reaches thermal equilibrium with the bath at $T = T_{\text{bath}} = 300 \text{ K}$. The external pressure is still

$$P_{\text{gas}} = P_{\text{atm}} + \frac{mg}{A} = 1.196 \times 10^5 \text{ Pa}.$$

Using the ideal gas law for the final equilibrium state:

$$V_{\text{final}} = \frac{nRT}{P_{\text{gas}}} = \frac{1 \times 8.314 \times 300}{1.196 \times 10^5} \approx \boxed{0.0209 \text{ m}^3}.$$

(d) Net heat

The initial and final temperatures are both 300 K, so over the entire process:

$$\Delta U_{\text{total}} = 0.$$

We split the process into two stages:

1. **Stage 1: Adiabatic compression** from V_1 to $V_{\text{eq,adiabatic}}$.
2. **Stage 2: Isobaric cooling** at constant $P_{\text{gas}} = 1.196 \times 10^5 \text{ Pa}$ from $(V_{\text{eq,adiabatic}}, T_{\text{eq}})$ to $(V_{\text{final}}, 300 \text{ K})$.

Stage 1 (adiabatic). From the adiabatic relation,

$$T_{\text{eq}} = T_1 \left(\frac{V_1}{V_{\text{eq,adiabatic}}} \right)^{\gamma-1} = 300 \left(\frac{0.0249}{0.0224} \right)^{2/3} \approx 322 \text{ K}.$$

Work done *by* the gas in the adiabatic stage:

$$W_1 = \frac{nR}{\gamma-1} (T_1 - T_{\text{eq}}) \approx \frac{8.314}{2/3} (300 - 322) \approx -2.8 \times 10^2 \text{ J}.$$

(negative: work is done *on* the gas).

Change in internal energy:

$$\Delta U_1 = nC_V (T_{\text{eq}} - T_1) = \frac{3}{2} nR (322 - 300) \approx 2.8 \times 10^2 \text{ J}.$$

Since the process is adiabatic, $Q_1 = 0$.

Stage 2 (isobaric cooling). In this stage, the gas cools from $T_{\text{eq}} \approx 322 \text{ K}$ to $T_{\text{final}} = 300 \text{ K}$ at constant pressure $P_{\text{gas}} = 1.196 \times 10^5 \text{ Pa}$.

Work done *by* the gas:

$$W_2 = P_{\text{gas}}(V_{\text{final}} - V_{\text{eq,adiabatic}}) \approx (1.196 \times 10^5) (0.0209 - 0.0224) \approx -1.8 \times 10^2 \text{ J}.$$

Change in internal energy:

$$\Delta U_2 = nC_V(T_{\text{final}} - T_{\text{eq}}) = \frac{3}{2}nR(300 - 322) \approx -2.8 \times 10^2 \text{ J}.$$

Using the first law with W = work done by the gas:

$$\Delta U_2 = Q_2 - W_2 \quad \Rightarrow \quad Q_2 = \Delta U_2 + W_2 \approx (-2.8 \times 10^2) + (-1.8 \times 10^2) \approx -4.6 \times 10^2 \text{ J}.$$

Total net heat. Over the whole process:

$$\Delta U_{\text{total}} = \Delta U_1 + \Delta U_2 = 0,$$

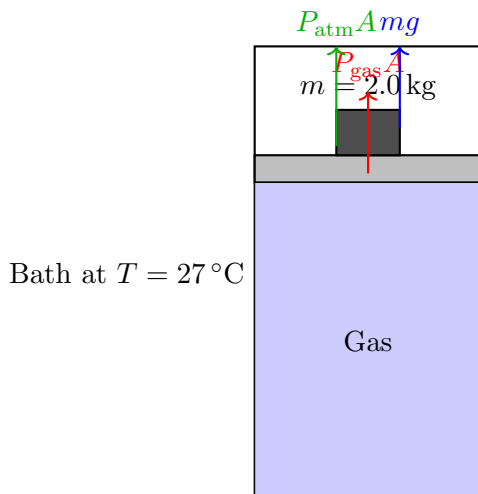
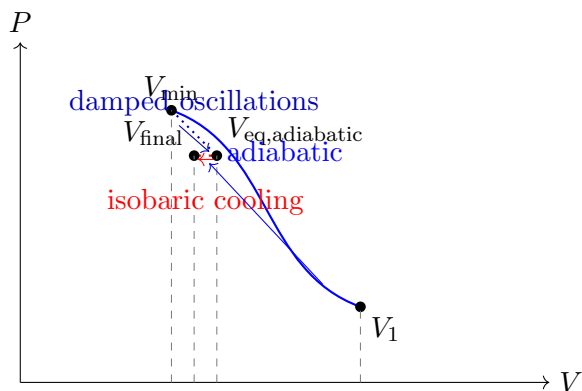
$$W_{\text{total}} = W_1 + W_2 \approx -4.6 \times 10^2 \text{ J},$$

$$Q_{\text{net}} = Q_1 + Q_2 \approx -4.6 \times 10^2 \text{ J}.$$

So the gas *releases* heat to the bath:

$Q_{\text{net}} \approx -4.6 \times 10^2 \text{ J (heat flows out of the gas)}.$

Required drawings:



Q9. A cycle with N₂ (3+3+2+3 pts)

Given: $n = 2 \text{ mol N}_2$ ($\gamma = 7/5$), $T_A = 300 \text{ K}$, $V_A = 0.0100 \text{ m}^3$, $V_B = 5 \text{ L} = 0.0050 \text{ m}^3$, $T_C = 300 \text{ K}$.

(a) Net work

Process A→B (adiabatic compression):

$$T_B = T_A \left(\frac{V_A}{V_B} \right)^{\gamma-1} = 300 \times \left(\frac{0.0100}{0.0050} \right)^{0.4} = 300 \times 2^{0.4} = 396 \text{ K}.$$

$$W_{AB} = \frac{nR}{\gamma-1} (T_A - T_B) = \frac{2 \times 8.314}{0.4} (300 - 396) = \boxed{-3.98 \times 10^3 \text{ J}}.$$

Process B→C (isochoric): $W_{BC} = 0$.

Process C→A (isothermal expansion):

$$W_{CA} = nRT_C \ln \frac{V_A}{V_C} = 2 \times 8.314 \times 300 \times \ln \frac{0.0100}{0.0050} = \boxed{3.46 \times 10^3 \text{ J}}.$$

$$\text{Net work: } W_{\text{net}} = W_{AB} + W_{BC} + W_{CA} = -3985 + 0 + 3458 = \boxed{-527 \text{ J}}.$$

(b) Net heat

$$\text{From first law: } Q_{\text{net}} = W_{\text{net}} = \boxed{-527 \text{ J}} \text{ (since } \Delta U_{\text{cycle}} = 0).$$

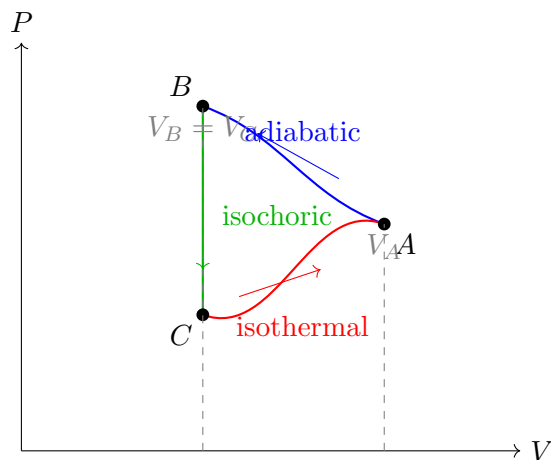
(c) Change in internal energy

$$\Delta U_{\text{cycle}} = \boxed{0}.$$

(d) Entropy change

For a complete cycle: $\Delta S_{\text{cycle}} = \boxed{0}$ (entropy is a state function).

Required drawing:



Q10. Carnot engine with N₂ (4+4+3+3+1+3 pts)

Given: $n = 1$ mol N₂, $T_h = 400$ K, $T_c = 300$ K, $V_a = 6.0$ L = 0.0060 m³, $V_c = 18.0$ L = 0.0180 m³, $\gamma = 7/5$.

(a) Volume V_b

At state b , isothermal expansion ends. For adiabatic process $b \rightarrow c$:

$$T_h V_b^{\gamma-1} = T_c V_c^{\gamma-1} \quad \Rightarrow \quad V_b = V_c \left(\frac{T_c}{T_h} \right)^{1/(\gamma-1)}.$$

$$V_b = 0.0180 \times \left(\frac{300}{400} \right)^{5/2} = 0.0180 \times 0.75^{2.5} = 0.0180 \times 0.4867 = \boxed{0.00877 \text{ m}^3 = 8.77 \text{ L}}.$$

(b) Volume V_d

At state d , adiabatic compression ends. For adiabatic process $d \rightarrow a$:

$$T_c V_d^{\gamma-1} = T_h V_a^{\gamma-1} \quad \Rightarrow \quad V_d = V_a \left(\frac{T_h}{T_c} \right)^{1/(\gamma-1)}.$$

$$V_d = 0.0060 \times \left(\frac{400}{300} \right)^{5/2} = 0.0060 \times 1.333^{2.5} = \boxed{0.0123 \text{ m}^3 = 12.3 \text{ L}}.$$

(c) Heat input Q_{in}

For isothermal expansion $a \rightarrow b$:

$$Q_{\text{in}} = nRT_h \ln \frac{V_b}{V_a} = 1 \times 8.314 \times 400 \times \ln \frac{0.00877}{0.0060} = \boxed{1.26 \times 10^3 \text{ J}}.$$

(d) Heat out Q_{out}

For isothermal compression $c \rightarrow d$:

$$Q_{\text{out}} = -nRT_c \ln \frac{V_d}{V_c} = -1 \times 8.314 \times 300 \times \ln \frac{0.0123}{0.0180} = \boxed{950 \text{ J}}.$$

(e) Efficiency

$$\eta = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{400} = \boxed{0.25 = 25\%}.$$

(f) Entropy change per gram

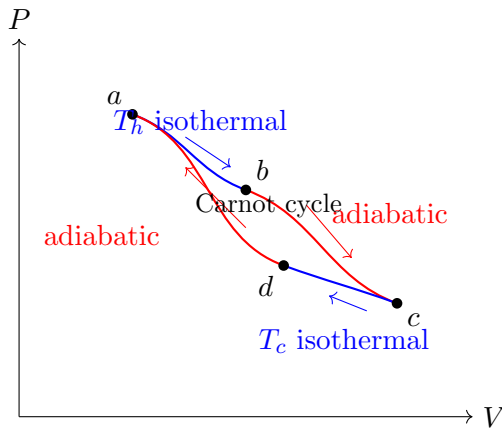
Molar mass of N₂: $M = 28$ g/mol.

Process $a \rightarrow b$ (isothermal): Using $V_b = 0.008769$ m³ (from part (a)):

$$\Delta S_{ab} = nR \ln \frac{V_b}{V_a} = 1 \times 8.314 \times \ln \frac{0.008769}{0.0060} = 3.154 \text{ J/K}.$$

Per gram: $\frac{\Delta S_{ab}}{M} = \frac{3.154}{28} = \boxed{0.113 \text{ J}/(\text{K} \cdot \text{g})}$.

Process b→c (adiabatic): $\Delta S_{bc} = 0$, so per gram: $\boxed{0}$.
Required drawing:



Bonus. Greenhouse effect (0–2 pts)

The greenhouse effect occurs when certain atmospheric gases (e.g., CO₂, water vapor, methane) absorb and re-emit infrared radiation emitted by Earth's surface, trapping heat and warming the planet. Without this effect, Earth's average temperature would be about 255 K, but the actual mean surface temperature is 288 K, demonstrating its importance.

To make Earth a better place to live:

1. Reduce greenhouse gas emissions by transitioning to renewable energy sources and improving energy efficiency.
2. Protect and restore natural carbon sinks like forests and oceans that absorb CO₂.
3. Develop and deploy carbon capture technologies to remove excess CO₂ from the atmosphere.

Assistance notes:

LLMs were used as translators and concept explainers. No direct solutions were provided.